

Software Test Plan

Project 27 | Project Management System | Version 1.0 | 28 September 2026

Prepared for the Software Engineering mini-project. Team and approval details may be completed by the submitting group.

1 Introduction

This plan specifies how the Project Management System requirements will be verified. The test scope includes account, project, task, board, effort, reporting, security and quality requirements. Results are recorded by case ID against SRS version 1.0.

2 Test items

Browser interface, JSON API, authorization middleware, project/task services, relational database, audit events and CSV export.

3 Features to be tested

SRS FR-01 through FR-17; NFR-01 through NFR-06; SR-01 through SR-05. Test data include three roles, two projects, tasks in all five statuses, due dates, and time entries.

4 Features not to be tested

External mail delivery, native mobile clients, billing and third-party integrations are outside version 1. Browser vendor internals and database engine internals are excluded.

5 Test approach and strategy

Unit tests validate rules and calculations. Integration tests cover API, permissions, persistence and audit. System tests exercise browser flows. Acceptance tests run against the SRS criteria. Use a clean seeded database for each case and record actual result, status, build and defect ID. Negative tests check rejected inputs and cross-project access. The performance run uses 500 tasks, 20 members, 20 concurrent users and 100 board/list requests, measuring server response times.

5.1 Security validation

TC-04 and TC-06 attempt unauthorized membership and task changes. TC-15 verifies salted password hashes, CSRF rejection, inert output for a script payload, and 15-minute lockout after five failures. Inspect server responses and stored records; do not rely only on hidden UI buttons. SR-01 through SR-05 are mandatory release checks.

6 Test environment

One application instance with a test relational database and current stable Chrome and Firefox. Use separate Owner, Manager, Member and outsider accounts. Reset fixture before each test; use a local load generator and timing logs for performance. Use test-only credentials and data.

7 Test schedule

Phase 1: case design and fixture setup. Phase 2: unit and integration execution. Phase 3: browser, security and load tests. Phase 4: defect fixes, regression and summary. Dates are assigned by the team after implementation milestones are agreed.

8 Test deliverables

Test case matrix, executable scripts or manual records, seeded data, execution log, defect list, requirement traceability, and a final test summary.

9 Roles and responsibilities

Role	Responsibility
QA lead	Own case coverage and results
Tester	Execute and log evidence
Developer	Fix defects and support diagnosis
Project owner	Accept release criteria

10 Risks and mitigation

Risk	Mitigation
Late stable build	Start service-level tests with early builds
Inconsistent fixture	Reset database before each run
Slow load environment	Record machine profile and isolate other workloads

11 Assumptions and dependencies

A test build, schema migration, seed fixture and test accounts are available. Clock and time zone behavior are configured consistently. Requirement changes trigger a traceability review.

12 Suspension and resumption criteria

Suspend affected testing if the database is unavailable, a blocking migration fails or more than 30% of cases cannot run. Resume after the environment is restored, a smoke test passes and fixture integrity is checked.

13 Test cases and traceability

Case	Scenario	SRS IDs	Procedure summary	Expected result
TC-01	Registration valid and duplicate	FR-01	Register fresh email; retry same email	One account; duplicate rejected
TC-02	Sign in and sign out	FR-02	Valid login; logout; invalid password	Session opens then closes; invalid denied
TC-03	Create project	FR-03	Submit complete project form	Project created; creator Owner
TC-04	Membership and role control	FR-04, SR-01	Owner invites; Member tries role change	Invite succeeds; Member denied
TC-05	Task creation validation	FR-05	Manager creates task; missing title attempt	Valid task in Backlog; invalid denied
TC-06	Task edit permissions	FR-06, SR-01	Assignee edits title; tries estimate; outsider accesses	Title saved; forbidden changes denied
TC-07	Kanban movement and audit	FR-07, FR-13	Move Backlog to In Progress	Board and history reflect transition
TC-08	Combined board filters	FR-08	Set assignee and priority filters	Intersection only
TC-09	Estimate bounds	FR-09	Submit 2.5, -1, 2.3 hours	2.5 accepted; others denied

TC-10	Time log calculation	FR-10, FR-11	Log 1.25 and 0.75; edit own entry	Actual hours recomputed correctly
TC-11	Dashboard totals	FR-11	Seed known status and overdue records	Counts and sums match fixture
TC-12	Comments and search scope	FR-12, FR-14	Comment and search across two projects	Comment ordered; cross-project result hidden
TC-13	CSV export and archive	FR-15, FR-16	Export; archive; try editing	CSV matches; archived edit denied
TC-14	Notification delivery	FR-17	Assign task and invite member	One notification per event
TC-15	Security handling	SR-02, SR-03, SR-04, SR-05	Inspect password store; CSRF omission; script payload; six failed logins	Hash only; CSRF rejected; script inert; lockout
TC-16	Board load performance	NFR-01, NFR-02	Seed 500 tasks, 20 members; 20 concurrent users, 100 requests	95th percentile ≤ 2 s; all records present
TC-17	Restart and rollback	NFR-03	Commit change, restart; inject DB failure	Committed value survives; no partial write
TC-18	Usability and browser checks	NFR-04, NFR-05, NFR-06	Count actions; keyboard walkthrough; Chrome and Firefox	≤ 10 actions; keyboard works; flows pass

14 Metrics and reporting

Report executed, passed, failed and blocked counts; requirement coverage (IDs with at least one executed case divided by total IDs); open defects by severity; and 95th percentile response time. Release gate: all high-priority and security cases pass, no critical defects remain, and each SRS ID is covered.

15 Approvals

Role	Name	Signature and date
QA lead		
Development lead		
Project owner		